The actual quadratic source correction in the last compact orbit

D.66. Two literal peripheral corrections fill the remaining old compact comparison block [Proved]

Retain the actual sources, arithmetic element, coefficient modules and marked quotient maps of D.65. In particular,

$$X = \widehat{B}/(R_{50} \cap L_2), \qquad Y = X/\mathcal{K}, \qquad X_{65} = X/\mathcal{J}_{65}, \qquad S_{57} = \mathfrak{R}_h \cap M_* \cap L_{*,2}. \quad \text{(D.66.1)}$$

The integral homology image of S_{57} is the actual lattice $\mathcal{L} = \ker(U^- \to U_*^-)$. The coefficient group remains the actual G of order $576 \cdot 3^{97}$, with its original source generators s, r and cusp $t = sr$ of order 36.

There is an actual continuous peripheral automorphism of

$$\begin{aligned} X_{66} &= X/\mathcal{J}_{66}, \\ \mathcal{J}_{66} &= \ker(\mathcal{J}_{63} \xrightarrow{(q_1,q_2,q_3,q_4)} C_1 \oplus C_2 \oplus C_3 \oplus C_4). \end{aligned} \quad \text{(D.66.2)}$$

It fixes the original cusp, has multiplier one, is identity on G, and retains the complete literal D.57 small action and the prescribed $\mathcal{W}_h$ action. All the proved normalized parameter, metabelian and CM comparisons through $\mathcal{W}_h$ are retained. Both arithmetic branches and every retained small framing are included. The homology action is the replacement described below.

The exact new projection kernel and dimensions are

$$\begin{aligned} \ker(X_{66} \to X_{65}) &= \mathcal{J}_{65}/\mathcal{J}_{66} \simeq \ker \nu_4, \\ \dim \ker \nu_4 &= 1918, & \dim(\ker \nu_4)_G &= 4, \\ \dim(\mathcal{J}_{63}/\mathcal{J}_{66}) &= 9600, & \dim(\mathcal{J}_{63}/\mathcal{J}_{66})_G &= 28, \\ \dim \mathcal{J}_{66} &= \dim \mathcal{J}_{65} - 1918, \\ \dim(\mathcal{J}_{66})_{\text{specified small characters}} &= 274560. \end{aligned} \quad \text{(D.66.3)}$$

Here ν_4 is the actual full norm used in D.65. The four-dimensional coinvariant space of its kernel maps with rank two to the coinvariants of C_4; both maps are retained.

We give the finite source detecting the new correction. On the old orbit $a = 4$, let M be the integral lattice induced from the actual rank-16 $\mathcal{O} = \mathbb{Z}_2[\omega]$-cover through the eight unordered character cosets. Its $\mathbb{Z}_2$-rank is 256. The actual small compact lattice M_* has rank 64. Write $\phi : \mathsf{M} \twoheadrightarrow \mathsf{M}_*$ for their marked map. There are actual marked epimorphisms

$$\begin{gathered} X \longrightarrow \mathscr{E} \longrightarrow \mathscr{E}_*, \\ \mathscr{E} = \mathscr{N} \rtimes G^\dagger, \quad |G^\dagger| = 46656, \quad \mathscr{N}^{\mathrm{ab}} = \mathsf{M}/4, \quad \mathscr{N}' = C \simeq \mathbb{F}_2^8, \\ \mathscr{E}_* = \mathscr{N}_* \rtimes G_*, \quad |G_*| = 3888, \quad \mathscr{N}_*^{\mathrm{ab}} = \mathsf{M}_*/4, \quad \mathscr{N}_*' = C_* \simeq \mathbb{F}_2^4. \end{gathered} \quad \text{(D.66.4)}$$

Both two-groups have class two and exponent four, and orders 2^{520} and 2^{132}. The second map induces the original base map and ϕ. Its central map has the retained columns

$$(\mathtt{2}, \mathtt{1}, \mathtt{4}, \mathtt{2}, \mathtt{8}, \mathtt{8}, \mathtt{1}, \mathtt{4}). \quad \text{(D.66.5)}$$

Typewriter masks denote hexadecimal binary-coordinate masks. The image of S_{57} in $\mathscr{N}$ is exactly $\ker(\mathscr{N} \to \mathscr{N}_*)$, with the integral homology and central lifts specified in the proof.

In this actual marked quotient the source cusp has zero $\mathsf{M}/4$-coordinate and central coordinate $c_t = \mathtt{26}$. Its base coordinate has order twelve. For $\ell(c) = c_4 + c_5$, with zero-based indices,

$$\ell(c_t) = 1, \qquad V_0 = \langle \mathtt{09}, \mathtt{42} \rangle \subset \ker(C \to C_*). \quad \text{(D.66.6)}$$

The image of $\mathcal{J}_{65}$ in C is exactly V_0. There are actual relative compact automorphisms whose central actions are all four maps

$$I + v\ell, \qquad v \in V_0. \tag{D.66.7}$$

They fix $\ker(C \to C_*)$ pointwise. Their two literal conjugator differences have zero image in the full small class-two quotient. Their cusp differences are respectively $\texttt{00}, \texttt{09}, \texttt{42}, \texttt{4b}$. The original source s has order two and r order three; all four corrections have zero effect on the displayed image of t^{18}.

The earlier integral test has an exact limitation. The secondary modulo-four norm on the two residual old compact coordinates is zero, as is the induced even-homology divided-orientation image. Both coordinates inject into the whole compact coinvariants. The new lift uses (D.66.7) and its literal quadratic cusp equation.

The full remaining source kernel is

$$\begin{aligned} \ker(X \to X_{66}) = \mathcal{J}_{66}, \qquad \ker(X_{66} \to Y) = \mathcal{K}/\mathcal{J}_{66}, \\ S_{66} = \{u \in R_{50} : \mathrm{im}_{\mathcal{T}_2}(u) \in \mathcal{J}_{66}\}, \\ S_{66}/(R_{50} \cap L_2) = \mathcal{J}_{66}. \end{aligned} \tag{D.66.8}$$

Its specified remaining central block has positive dimension 274560. The remaining arithmetic boundary and the full profinite realization of D.13/D.19 are not determined here. The manuscript passages are Part D(e), section 51.16, folio 734 verso, scan 174, and section 51.15, folios 725–727, scans 156–160.

Proof. Integral marked projective lattices. Use the old character $(0, 1, 0, 0)$, its unordered stabilizer H of order 72, and the actual graph of D.60. The affine, cusp and compact modules on this orbit are projective over the actual integral coefficient group, by D.60.12 and its integral lifting argument. The old compact cover is a projective direct summand. The small compact module is projective over its actual small coefficient group as well.

The program `d_old_compact_integral.py` retains the D.64 cycles modulo four and their original reduction. In the real 32-dimensional cover module it chooses four generators for the free Sylow-eight restriction. The 32 translated columns form a basis. The coefficient-of-identity rows give an ordinary map whose Sylow trace is the identity. Averaging over the odd scalar group and the remaining stabilizer cosets lifts the retained projector. The polynomial $3e^2 - 2e^3$ makes it idempotent modulo four. The rank-16 cover, all 72 semilinear actions, and the actual rank-four small map retain their D.64 reductions. The small map kills the rank-32 complementary projective summand exactly.

These maps lift integrally. Idempotents lift through successive powers of two; projectivity lifts the prescribed inclusion and retraction, as in D.61. The actual graph and cusp kernels are saturated. Applying the projective-domain Hom functor lifts their specified comparison maps and reductions. The odd quotient from G to $G^\dagger$ is the actual D.60.10 map, with kernel of order 3^{93}. Its normalized transfer is the inverse lattice comparison, with odd denominators. Thus M, its graph projection and its small map are actual integral summands and maps of the full arithmetic homology.

The complete compact central comparisons. On one unordered pair the odd-invariant exterior module is the D.64 Hermitian module. In the real basis $u_i, \omega u_i$, use exactly the diagonal and off-diagonal exterior identifications of D.64.12. Its complete stabilizer coinvariant quotient has dimension eight. The actual small Hermitian quotient has dimension four. Applying $B \mapsto \phi B \bar{\phi}^t$ to all 256 Hermitian basis vectors gives (D.66.5).

The comparison-kernel coinvariants inject into the whole coinvariant quotient, with columns

$$\texttt{09}, \quad \texttt{30}, \quad \texttt{42}, \quad \texttt{84}. \tag{D.66.9}$$

They are exactly the kernel of (D.66.5). The ordinary norm has rank two on this space; its kernel there is V_0. This verifies the central comparison through the actual exterior map, rather than a dimension count alone.

The full induced Q-module has dimension 1920. The program `d_old_compact_global_kernel.py` constructs its source s, r matrices through the same eight coset representatives and complete stabilizer actions. It checks both generator relations. The full norm has rank two. An actual basis of its 1918-dimensional kernel gives 3836 augmentation columns of rank 1914. Its coinvariant dimension is therefore four. Its further map into the eight-dimensional whole compact quotient has rank two and image exactly V_0. The odd parameter subgroup acts trivially on the averaged self module, with odd norm factor. These are also the actual G-module maps.

The secondary integral test is kept separately. Lift the 240-dimensional comparison kernel into the actual integral Hermitian kernel modulo four by subtracting a right-inverse image for ϕ. On an ordinary-norm-kernel vector, divide the full signed integral norm by two, then reduce modulo the ordinary norm image. A changed lift adds an ordinary norm. The program `d_old_compact_secondary.py` checks all augmentation relations and both residual generators; their secondary images are zero. The even endomorphism trace also gives zero divided-orientation image in both branches. The injection (D.66.9) proves that both coordinates survive whole compact augmentation. These computations are the asserted limitations.

The actual projective edge extension. Let $E = \mathbb{Z}_2[G]$ be the Bass–Serre edge module and let $p : U \to \mathsf{M}$ be the actual compact projection. The coefficient-of-identity construction for the projective target extends p to an equivariant map $a : E \to \mathsf{M}$. On the selected character fibre, extend the coefficient functional ordinarily from cycles to edges, sum its Sylow translates, and average over the odd remaining cosets. The cycle and edge quotient lattices are free over $\mathbb{Z}_2$, so the ordinary extension exists. The same formula is integral relative injectivity and allows the prescribed modulo-four extension. The scalar average has coefficient $1/3$, and the stabilizer relative trace has coefficient $1/[H : \mathrm{Syl}_2 H] = 1/9$. Modulo four these coefficients are three and one, exactly as in the program. The reduction lifting can also be seen from $H^1(G, \mathrm{Hom}_{\mathbb{Z}_2}(E/U, \mathsf{M})) = 0$: this Hom lattice is a summand of an induced lattice, with the literal twisting map $g \otimes v \mapsto g \otimes g^{-1}v$.

Write $w = a(e_1)$. The actual path cocycle is $p_B(s) = 0$, $p_B(r) = e_1 - e_r$, whence

$$ap_B(s) = 0, \qquad ap_B(r) = (1 - r)w, \qquad ap_B(t) = (s - t)w. \tag{D.66.10}$$

The compact projection kills every actual cusp cycle. Thus $N_{36}(t)(s - t)w = 0$. The lattice action factors through the cusp of order twelve in $G^\dagger$, so division by three gives the same norm equation there. Projectivity over the cyclic subgroup gives $\ker N_{12} = (t - 1)\mathsf{M}$. Choose

$$(t - 1)b = (s - t)w, \qquad \eta_s = (1 - s)b, \qquad \eta_r = (1 - r)(w + b), \qquad \eta_t = 0. \tag{D.66.11}$$

The specified solution modulo four lifts integrally: the difference of two reduced solutions is t-fixed; fixed vectors in a projective cyclic lattice are norm vectors, whose reductions lift. This proves compatibility at every integral homology level.

The program `d_old_compact_source_cochain.py` restricts its extension to exactly the retained 96-cycle-to-16-cover map and checks all 72 stabilizer equations. It induces the full rank-256 real module. The matrix of $t - 1$ has rank 232. The retained solution and complete source matrices satisfy (D.66.11) modulo four exactly.

A class-two group with the actual exterior map. In one real character fibre put

$$q(v) = [v\bar{v}^{\,t}] \in C, \qquad c(v, w) = [v\bar{w}^{\,t} + w\bar{v}^{\,t}], \tag{D.66.12}$$

where the brackets denote the actual Hermitian coinvariant map. The polarization of q is c. Start with the bilinear map whose diagonal is $q(e_i)$, whose strict lower triangle is $c(e_i, e_j)$, and

whose strict upper triangle is zero. Average its scalar translates by $1, \omega, \omega^2$. Since three is one modulo two, the resulting map B has the same diagonal and polarization and is scalar invariant. Sum over the eight unordered fibres and define

$$(v, c)(w, d) = (v + w, c + d + B(\bar{v}, \bar{w})), \qquad v, w \in \mathsf{M}/4, \quad c, d \in C. \tag{D.66.13}$$

Here bars on v, w mean reduction modulo two. Bilinearity is the associativity identity. The commutator is $c(\bar{v}, \bar{w})$, and the square is $(2v, q(\bar{v}))$, so the fourth power is one. The exterior images span C. Hence the derived group is C and the abelianization is $\mathsf{M}/4$.

For a deck matrix A_g, the bilinear difference $B(A_g v, A_g w) + B(v, w)$ is symmetric with zero diagonal. Take its triangular quadratic primitive and average the scalar translates. The resulting function k_g gives an automorphism

$$g(v, c) = (A_g v, c + k_g(\bar{v})). \tag{D.66.14}$$

The polar identity proves this assertion. Two such scalar-equivariant corrections differ by an additive map from a nontrivial odd-character module to a trivial module, so are equal. This proves all deck composition identities, including character exchange and permutation of fibres. The same construction and uniqueness apply to the actual small map and to any equivariant automorphism with its induced central map.

In $\mathscr{N} \rtimes G^\dagger$ put

$$S = \mathrm{Int}((b, 0))(s), \qquad R = \mathrm{Int}((w + b, 0))(r). \tag{D.66.15}$$

Their orders are two and three, as literal conjugates of the marked finite generators. Their product has zero abelian coordinate by (D.66.11). The complete quadratic evaluation gives

$$SR = (0, \mathsf{26}; t). \tag{D.66.16}$$

Its eighteenth power has central coordinate zero and base coordinate t^6. In the actual affine base this is the cusp involution retained in D.65, with its separate source marking.

The universal property of the profinite free product gives a map from $\widehat{B}$. On the base kernel its homology map is the actual onto projection p modulo two. Finite Frattini generation makes the image the whole $\mathscr{N}$, proving surjectivity. Since K_3 maps into a class-two exponent-four group, the map kills L_2 and factors through X.

The actual ϕ and (D.66.5) similarly give a unique scalar-equivariant class-two map to the small group. It commutes with all deck actions by uniqueness, and is onto by its Frattini map. Together with $G^\dagger \to G_*$ it gives $\mathscr{E} \to \mathscr{E}_*$. The original K_* maps into $\mathscr{N}_*$, so the composed source map kills $L_{*,2}$. This proves the comparison with the actual small class-two source.

All isometries and the relative quadratic correction. The program `d_old_compact_all_isometries.py` enumerates the complete affine spaces of D.64.17 and tests the entire isometry equations. On this orbit there are six binary parameters, hence 64 candidates, in each branch. Exactly 16 are isometries in each branch. Relative to the D.64 chosen solution X_0, put $\delta = X X_0^{-1}$. Then

$$\phi\delta = \phi, \qquad \delta^t H \bar{\delta} = H. \tag{D.66.17}$$

The full induced central actions are (D.66.7), each four times. The functional ℓ is the last coordinate of (D.66.5); it vanishes on that map's kernel. The vectors $\mathtt{09}, \mathtt{42}$ are in this kernel. This proves the claimed pointwise fixedness.

Every such matrix has an actual integral equivariant lift with $\phi\delta = \phi$. The reduced $\delta - I$ maps to the saturated lattice $\ker \phi$, and its domain is projective, so the equivariant Hom reduction is onto. The certificate retains a specific lift modulo four: project an ordinary lift into $\ker \phi$, then apply the coefficient-of-identity trace from the projective domain. Its reduction,

all 72 equivariance equations and its exact small square are checked. The integral relative average includes $1/[H : \mathrm{Syl}_2 H] = 1/9$; this scalar is one modulo four. Scalar averaging includes $1/3$, whose reduction modulo four is three. Projectivity lifts the specified modulo-four map integrally. Invertibility modulo two makes it an integral automorphism. The reduced isometry identity (D.66.17) is the one used in the present central equation.

Let F_δ be its canonical automorphism of $\mathscr{N}$. It commutes with the full deck action. Its composition with the small class-two map equals that map: both the linear and central squares commute, and uniqueness of the scalar-equivariant primitive proves the full square. Define

$$\begin{aligned} d_s &= F_\delta((b,0))(b,0)^{-1}, \\ d_r &= F_\delta((w+b,0))(w+b,0)^{-1}. \end{aligned} \tag{D.66.18}$$

Both have zero small class-two image, including their central coordinates. Conjugating (D.66.15) by them gives its images under F_δ. Consequently the literal new cusp differs by

$$(\delta_C - I)c_t = v\ell(c_t) = v. \tag{D.66.19}$$

The programs evaluate both source conjugations and their product for all 32 cases, with every linear and quadratic small comparison coefficient. The source orders remain exact. The eighteenth power is unchanged because its central difference is eighteen times an element of order two. This is the image of the original t^{18} through the stated quotient; the full source G still has cusp order 36.

The exact source intersection and integral lift. Let $P = U/\mathcal{R}$ be actual compactification, with small cusp lattice $\mathcal{R}_*$. D.64.11 proves the actual onto maps

$$\mathcal{R} \twoheadrightarrow \mathcal{R}_*, \qquad \mathcal{L} \twoheadrightarrow \ker(P \to P_*). \tag{D.66.20}$$

Their cusp width factor is three, a unit over $\mathbb{Z}_2$. Choose an equivariant affine section σ_* of the included small compact summand M_*. It exists by projectivity over its actual small group. Projectivity of M lifts its compact inclusion to an affine map. Its difference from $\sigma_*\phi$ after the small map lies in $\mathcal{R}_*$. Lift this difference through the first map in (D.66.20) and subtract it. This gives actual integral maps

$$\begin{gathered} \sigma : \mathsf{M} \longrightarrow U, \qquad p\sigma = I, \qquad \theta_U \sigma = \sigma_* \phi, \\ \sigma(\ker \phi) \subset \mathcal{L}. \end{gathered} \tag{D.66.21}$$

They use the original compact inclusion and projection. The subtracted cusp term is killed by p, so these equations hold integrally with precisely the finite matrices already used.

D.58 identifies the integral homology image of S_{57} with $\mathcal{L}$. The subgroup $R_{50} \cap L_{*,2}$ has image $\mathcal{K}$ in the central layer of $\mathcal{T}_2$, by the two actual source inclusions after D.58.16. D.64's surjective compact kernel map and (D.66.9) thus give

$$\mathrm{im}_C(R_{50} \cap L_{*,2}) = \ker(C \to C_*). \tag{D.66.22}$$

On these source elements the $\mathscr{N}$-homology is zero, and the central map is exactly the calculated exterior map. This proves the surjectivity with its actual source subgroup.

Equations (D.66.20)–(D.66.22) prove both inclusions in

$$\mathrm{im}_{\mathscr{N}}(S_{57}) = \ker(\mathscr{N} \to \mathscr{N}_*). \tag{D.66.23}$$

One follows from $S_{57} \subset L_{*,2}$ and the already proved factorization of the small source map. For the other, lift a prescribed linear coordinate in $\ker \phi$ through (D.66.21) and $S_{57} \to \mathcal{L}$. The

remaining discrepancy is central and is removed by (D.66.22). This also permits a prescribed full integral homology vector with the chosen projected linear coordinate.

Use the integral source homology conjugators

$$\gamma_s = \sigma((\delta - I)b), \qquad \gamma_r = \sigma((\delta - I)(w + b)). \tag{D.66.24}$$

They belong to $\mathcal{L}$. Their source cochain is $\eta = \sigma(\delta - I)\eta^{\mathsf{M}}$, where η^{M} is (D.66.11). Its value on t is zero integrally. The homology map is $I + \sigma(\delta - I)p$; it is identity on $\ker p$ and δ on $\sigma(\mathsf{M})$. Hence it is an automorphism at every integral level.

Lift (D.66.24) to two elements of S_{57}. Correct their central coordinates by (D.66.22) so that their images are exactly (D.66.18). This retains their entire integral homology. The resulting source map δ_B has the literal commuting square

$$q_E \delta_B = F_\delta q_E : X \longrightarrow \mathscr{E}. \tag{D.66.25}$$

The equation holds on both marked generators and thus on the source. No descent of the preliminary D.65 action to $\mathscr{E}$ is used.

Here is preservation of the exact source kernel. The two conjugators lie in $S_{57} \subset \mathfrak{R}_h$, so the map is identity on $\mathcal{W}_h$. It is identity on G, preserves K_3, and preserves its characteristic homology and class-two kernels M_∞, L_2. Through the $\mathcal{W}_h$ map it preserves $\mathfrak{R}_h$. It therefore preserves $R_{50} \cap L_2$ and factors through X. More explicitly, each of the three kernels $\mathfrak{R}_h, M_\infty, L_2$ contains the defining kernel of X; the intersection of their three images in X is trivial. The source homomorphism therefore kills their original intersection, without any assumption of an automorphism of $\widehat{B}$. It is identity on the complete small class-two source because both conjugators are in $L_{*,2}$. Its homology map is invertible, and its finite class-two map is invertible by the Frattini criterion. The exterior map preserves the comparison kernel and is invertible there. The exact marked fibre-product description of D.58 now gives an automorphism of X. It fixes t on Y, with a possible error in $\mathcal{K}$ on X.

The normalized parameter comparison is retained through its actual source map. D.49 gives $\mathfrak{R}_h \subset C_h \subset \widehat{F}'$. All correcting conjugators lie in $S_{57} \subset \mathfrak{R}_h$, so their abelian coordinates in $\widehat{F}$ are zero. The existing common commutator gauge and its literal image f_h^* in $\mathcal{W}_h$ are preserved.

Removal of the two actual residual errors. Take a preliminary D.65 lift to X with cusp error $z \in \mathcal{J}_{65}$, before quotienting that subgroup. Its actual conjugators preserve the same source kernels and give a source automorphism with that central error. Its image in C lies in V_0, by the exact norm-kernel map following (D.66.9).

The automorphism δ_B may have errors in previously controlled components. Repair them by the already proved source procedures of D.59–D.65. These preserve (D.66.25), as follows. The extra-projective and cusp corrections have zero M-homology, hence central images in $\mathscr{E}$; their source conjugations act trivially there. The same applies to corrections supported on the other three character orbits. Central two-conjugator augmentation corrections also have zero effect in C. The changed compact matrix is an isometry modulo two and is identity on the orthogonal extra-projective summands. The actual orientation and injective-norm arguments therefore restore the earlier cross and self conditions. The fourth compact ordinary norm is zero by the same orientation identity. These operations put its error in $\mathcal{J}_{65}$ while retaining (D.66.25). The source-involution words are followed by the actual moving augmentation repair, in the order proved in D.64 and D.65.

Choose $v = q_E(z)$ in (D.66.7). The correction preserves $\mathcal{J}_{65}$: the first three compact kernels remain invariant, it preserves the fourth comparison kernel through the actual small square $\phi\delta = \phi$, and its G-equivariant exterior map commutes with the full norm ν_4. The repaired

correction error and the transported old error therefore both remain in $\mathcal{J}_{65}$. The composite error has the exact image

$$\begin{aligned} q_E\big(\delta_B \alpha_{65}(t) t^{-1}\big) &= \delta_C(q_E(z)) + (\delta_C - I)c_t \\ &= q_E(z) + v = 0. \end{aligned} \tag{D.66.26}$$

The first equality uses only (D.66.25). The second uses pointwise fixedness on the comparison kernel and $\ell(c_t) = 1$. This is an actual source error calculation.

The remaining fourth compact error has zero coinvariant. Apply the actual augmentation map $(b, d) \mapsto (1 - s)b + s(1 - r)d$ to remove it. The four compact maps from $\mathcal{J}_{63}$ are jointly onto by D.64, so choose its coefficients with the other three compact coordinates zero. Their source lifts are in $R_{50} \cap L_{*,2}$. They preserve full homology, the literal small conjugators and $\mathcal{W}_h$. The remaining error is exactly in $\mathcal{J}_{66}$.

On Y, the replacement is the automorphism of (D.66.24), followed by the D.65 action and the retained extra-projective cusp-inner repairs:

$$\mathrm{Int}(b_{t,Y})\delta_Y \alpha_{65,Y}, \qquad b_t = \prod_j \prod_{i=0}^{35} t^i u_j t^{-i}, \quad u_j \in S_{57}. \tag{D.66.27}$$

Every one of the 36 factors is retained. Its Y-image centralizes t; its small and $\mathcal{W}_h$ images are zero. The subsequent source-involution and central augmentation corrections leave this Y-action unchanged. This is the exact comparison with the previous representative.

The defining kernel $\mathcal{J}_{66}$ is preserved: compact projections kill cusp terms, the extra left rows are retained, all four compact reductions are invertible, and their actual small squares commute. The finite class-two map is an automorphism by its Frattini map. The same literal source conjugators give all integral homology levels and retain the prescribed $\mathcal{W}_h$ action. The actual fibre-product maps therefore prove (D.66.2).

The remaining kernel and execution scope. Joint surjectivity of the compact kernel maps gives $\mathcal{J}_{65}/\mathcal{J}_{66} \simeq \ker \nu_4$, with the original source projection. The computed dimensions give (D.66.3): $7682 + 1918 = 9600$, $4 + 10 + 10 + 4 = 28$, and $276478 - 1918 = 274560$. The positive remaining specified block proves that (D.66.8) is a proper nonzero source kernel.

Nine complete programs retain the actual integral lattices, all norm and quotient columns, complete isometry families, the projective edge extension, full quadratic group maps, relative small comparisons and induced norm-kernel module. Their actual outputs and immutable input hashes accompany the proof. The earlier missing-helper and syntax failures are preserved with their actual transcripts, as is the successful norm-kernel run before its extra central comparison was added. Only the successful completed executions support the assertions here. □